Formula List for Electrodynamics

1. Divergence of a vector

$$(\vec{\nabla}.\vec{A}) = \hat{\imath}\frac{\partial}{\partial x}A_x + \hat{\jmath}\frac{\partial}{\partial y}A_y + \hat{k}\frac{\partial}{\partial z}A_z$$

Where;

- $\vec{\nabla} = \hat{\imath}\frac{\partial}{\partial x} + \hat{\jmath}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$ is the del operator
- $\vec{A}$= Any arbitrary vector
- $\vec{\nabla}.\vec{A}$= divergence of vector A

2. Curl of a vector

$$(\vec{\nabla} \times \vec{A}) = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

Where;

- $\vec{\nabla} = \hat{\imath}\frac{\partial}{\partial x} + \hat{\jmath}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$ is the del operator
- $\vec{A}$= Any arbitrary vector
- $(\vec{\nabla} \times \vec{A})$= curl of vector A

3. Gradient of a Scalar function

$$\vec{\nabla}f(x,y,z) = \hat{\imath}\frac{\partial}{\partial x}f(x,y,z) + \hat{\jmath}\frac{\partial}{\partial y}f(x,y,z) + \hat{k}\frac{\partial}{\partial z}f(x,y,z)$$

Where;

- $\vec{\nabla} = \hat{\imath}\frac{\partial}{\partial x} + \hat{\jmath}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$ is the del operator
- $f(x,y,z)$= Any arbitrary function of x, y and z.
- $\vec{\nabla} f(x,y,z)$= Gradient of a function f(x,y,z)

4. Conversion of Cartesian co-ordinates to cylindrical

$$r = \sqrt{x^2 + y^2}$$
$$\emptyset = tan^{-1}\left(\frac{y}{x}\right)$$
$$z = z$$

Where;

- $r, \emptyset, z$ = The Cylindrical co-ordinates
- x, y, z = The Cartesian co-ordinates

5. Conversion of Cartesian co-ordinates to Spherical

$$r = \sqrt{x^2 + y^2 + z^2}$$
$$\theta = \tan^{-1}\left(\frac{\sqrt{x^2 + y^2}}{z}\right)$$
$$\emptyset = \tan^{-1}\left(\frac{y}{x}\right)$$

Where;

- $r, \theta, \emptyset$ = The Spherical co-ordinates
- x, y, z = The Cartesian co-ordinates